

1.2 Atoms, Molecules & Stoichiometry

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.2 Atoms, Molecules & Stoichiometry
Difficulty	Medium

Time allowed: 20

Score: /10

Percentage: /100

Question 1

Use of the data booklet is relevant to this question

A hydrated carbonate of an unknown Group 1 metal has the formula $X_2CO_3 \cdot 10H_2O$ and is found to have a relative formula mass of 286.

What is the group 1 metal?

- A. Rb
- B. K
- C. Na
- D. Li

[1 mark]

Question 2

Use of the data booklet is relevant to this question

A volume of nitrogen gas, 2.00 dm^3 measured under standard conditions, was reacted with a large volume of hydrogen gas, to produce ammonia. Only 15.0% of the nitrogen gas reacted to produce ammonia.

What mass of ammonia was made?

- A. 2.83 g
- B. 1.42 g
- C. 0.455 g
- D. 0.213 g

[1 mark]

Question 3

Use of the data booklet is relevant to this question.

Excess aqueous cold sodium hydroxide is reacted with 0.125 mol of chlorine gas, Cl_2 . One of the products is a compound of sodium, oxygen and chlorine.

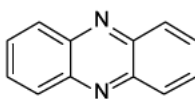
What mass of the product is formed?

- A. 13.3 g
- B. 9.31 g
- C. 18.6 g
- D. 26.6 g

[1 mark]

Question 4

The diagram below shows the skeletal formula of phenazine.



Phenazine

What is the empirical formula of phenazine?

- A. $\text{C}_6\text{H}_6\text{N}$
- B. $\text{C}_{12}\text{H}_8\text{N}_2$
- C. $\text{C}_6\text{H}_4\text{N}$
- D. $\text{C}_{12}\text{H}_{12}\text{N}_2$

[1 mark]

Question 5

Use of the data booklet is relevant to this question.

When heated magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, will decompose to a white solid and a mixture of gases. One of the gases released is an oxide of nitrogen, X.

7.4 g of anhydrous magnesium nitrate is heated until no further reaction takes place.

What is the mass of X produced?

- A. 4.6 g
- B. 1.5 g
- C. 2.3 g
- D. 3.0 g

[1 mark]

Question 6

Use of the data booklet is relevant to this question.

Excess water and 0.144 g of an aluminium compound X were reacted to form a gas. The gas burns completely in O_2 to form H_2O and 72 cm^3 of CO_2 only. The volume of CO_2 was measured at room temperature and pressure.

What is the formula for X?

- A. Al_2C_3
- B. Al_3C_4
- C. Al_5C_3
- D. Al_4C_3

[1 mark]

Question 7

Use of the data booklet is relevant to this question.

A metal element of mass 1.15 g reacted with 300 cm³ of oxygen at 298 K and 1 atm pressure, to form an oxide which contains O²⁻ ions.

Which one of these is the metal?

- A. Calcium
- B. Sodium
- C. Potassium
- D. Magnesium

[1 mark]

Question 8

Use of the data booklet is relevant to this question

The mass of a coin is 20% nickel. The coin has a mass of 10.0 g.

How many nickel atoms are in the coin?

- A. 4.30×10^{22}
- B. 1.03×10^{23}
- C. 1.20×10^{24}
- D. 2.05×10^{22}

[1 mark]

Question 9

Use of the data booklet is relevant to this question.

An additive in leaded petrol is composed of lead, carbon and hydrogen only. The compound contains 6.19% hydrogen and 29.7% carbon by mass.

What is the value of x in the empirical formula PbC_8H_x ?

- A. 6
- B. 5
- C. 20
- D. 16

[1 mark]

Question 10

Use of the data booklet is relevant to this question

Solid fertilisers often contain the elements N, P and K in a ratio of 15 g: 30 g: 15 g per 100 g of fertiliser. It is recommended that the fertiliser is used at 14 g of fertiliser per 5 dm^3 of water.

What is the concentration of nitrogen atoms in the solution?

- A. 0.42 mol dm^{-3}
- B. 0.05 mol dm^{-3}
- C. 0.03 mol dm^{-3}
- D. 0.75 mol dm^{-3}

[1 mark]

Model Answers: Medium

1

The correct answer is **C** because:

Steps		Calculation
1	Create an equation that links the atomic masses of each atom present to the overall relative formula mass	$2X + 12 + 48 + 20 + (10 \times 16) = 286$
2	Rearrange to find X	$2x + 240 = 286$ $X = 23$
3	Find the element that has an atomic mass of 23	Sodium (Na)

2

The correct answer is **C** because:

Steps		Calculation
1	Calculate the moles of the gas using the formula: $\text{Amount of gas (mol)} = \frac{\text{Volume of gas (dm}^3\text{)}}{22.4}$	$2 / 22.4 = 0.0893$
2	Only 15% of the nitrogen used	$0.0893 \times 0.15 = 0.0134 \text{ g}$
3	Mass of Ammonia $\text{Moles} = \text{mass} / M_r$ Re-arrange for mass $\text{Mass} = \text{moles} \times M_r$	$0.0134 \times 17 = 0.2277 \text{ g}$

4	Nitrogen is a diatomic molecule so for each molecule of nitrogen 2 ammonia molecules will be produced	$0.2277 \times 2 = \mathbf{0.455\text{ g}}$
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3

The correct answer is **B** because:

Steps		Calculation
1	Balanced symbol equation for the cold reaction	$2\text{NaOH (aq)} + \text{Cl}_2(\text{g}) \rightarrow \text{NaOCl}(\text{aq}) + \text{H}_2\text{O (l)} + \text{NaCl}(\text{aq})$
2	Calculate the relative formula mass of the compound M_r	$23 + 16 + 35.5 = 74.5$
3	Mass of Compound $\text{Moles} = \text{mass} / M_r$ Re-arrange for mass $\text{Mass} = \text{moles} \times M_r$	$0.125 \times 74.5 = \mathbf{9.31\text{ g}}$

4

The correct answer is **C** because:

- The **empirical formula** is the simplest whole number ratio of atoms in a compound.
- The formula of phenazine is $(\text{C}_6\text{H}_4)_2\text{N}_2$.
- The simplest ratio of the atoms would be $\text{C}_6\text{H}_4\text{N}$.

A is incorrect as the ratio of hydrogen atoms is not 6.

B is incorrect as this is the **molecular formula**.

D is incorrect as the number of hydrogen is incorrect.

5

The correct answer is **A** because:

Steps		Calculation
1	Balanced symbol equation for the reaction	$2\text{Mg}(\text{NO}_3)_2 (\text{s}) \rightarrow 2\text{MgO} (\text{s}) + 4\text{NO}_2 (\text{g}) + \text{O}_2 (\text{g})$
2	Calculate the relative formula mass (M_r) of compound $\text{Mg}(\text{NO}_3)_2$	$24.3 + (14 \times 2) + (16 \times 6) = 148.3$
3	Moles of $\text{Mg}(\text{NO}_3)_2$ $\text{Moles} = \text{mass} / M_r$	Moles = $7.4 / 148.3$ $= 0.0498$
4	Ratio of moles in equation: 2 moles of $\text{Mg}(\text{NO}_3)_2$ produces 4 moles of NO_2	$0.0498 \times 2 = 0.0998$
5	Mass of NO_2 $\text{Moles} = \text{mass} / M_r$ Re-arrange for mass $\text{Mass} = \text{moles} \times M_r$	$0.0998 \times 46 = 4.5908 \text{ g}$ 1dp = 4.6 g

6

The correct answer is **D** because:

Steps		Calculation
1	Balanced symbol equations for the reaction. This is a complete combustion reaction producing H_2O and CO_2	$\text{Al}_x\text{C}_y + \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{CH}_4$ $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$
2	72cm^3 is equivalent to 0.003 moles of carbon dioxide	$0.072 / 24 = 0.003$

$$\text{Amount of gas (mol)} = \frac{\text{Volume of gas (dm}^3\text{)}}{24}$$

5 $Moles = mass / M_r$ Re-arrange for mass $Mass = moles \times M_r$	$0.0998 \times 46 = 4.5908 \text{ g}$ 1dp = 4.6 g
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6

The correct answer is **D** because:

Steps	Calculation
1 Balanced symbol equations for the reaction. This is a complete combustion reaction producing H_2O and CO_2	$\text{Al}_x\text{C}_y + \text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{CH}_4$ $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{CO}_2$
2 72cm^3 is equivalent to 0.003 moles of carbon dioxide $Amount\ of\ gas\ (mol) = \frac{Volume\ of\ gas\ (dm^3)}{24}$	$0.072 / 24 = 0.003$
3 Find the total mass of Carbon	$0.003 \times 12 = 0.036 \text{ g}$
4 Subtract 0.036 g from the starting mass as the mass of carbon at the end is the mass of the carbon at the start	$0.144 - 0.036 = 0.108 \text{ g}$
5 Moles of Al $Moles = mass / M_r$	$0.108 / 27 = 0.004$ moles
6 Ratio of Al to C. Al has a valency of 3+ and C 4- this ratio also leaves an electrically balanced compound	Al = 4 C = 3

7

The correct answer is **B** because:

Steps		Calculation
1	300cm³ is equivalent to 0.0125 moles of oxygen Amount of gas (mol) = $\frac{\text{Volume of gas (dm}^3\text{)}}{24}$	0.3 / 24 = 0.0125
2	There are two options for the metal either group 1 metal or group 2 metal. Group 1 metals - 4 metal atoms would react with each oxygen atom. Group 2 metals - 2 metal atoms would react with each oxygen atom	Group 2 metals: $2M + O_2 \rightarrow 2MO$ Group 1 metals: $4M + O_2 \rightarrow 2M_2O$
3	M_r of the metal $M_r = \text{mass} / \text{moles}$ The metal from the list with an M_r that matches one of the possible calculations is sodium	Group 2 metal $1.15 / (0.0125 \times 2) = 46$ Or Group 1 metal $1.15 / (0.0125 \times 4) = 23$

8

The correct answer is **D** because:

Steps		Calculation
1	Find the mass of the Nickel in the coin	10 x 0.2 = 2 g
2	Find the moles of Nickel in the coin $\text{Moles} = \text{mass} / M_r$	2 / 58.7 = 0.034
	Find the number of atoms in the coin	

3	6.022×10^{23} atoms per mole (mol^{-1}) – this is known as Avogadro number , this has the symbol <i>L</i>	$0.034 \times (6.022 \times 10^{23}) = 2.05 \times 10^{22}$
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The correct answer is **C** because:

	Steps	Calculation
1	Find the moles of carbon and hydrogen, assume you have 100g so the percentages are equal to mass $Moles = mass / M_r$	Carbon $Moles = 29.7 / 12 = 2.475$ Hydrogen $Moles = 6.19 / 1 = 6.19$
2	Divide both side by the smallest number	Carbon $2.475 / 2.475 = 1$ Hydrogen $6.19 / 2.475 = 2.5$
3	Multiply by the number of carbon atoms given in the question	$2.5 \times 8 = 20$ The number of hydrogen atoms is 20

10

The correct answer is **C** because:

Steps	Calculation

1 **Mass** of nitrogen in 14 g of fertiliser

$$15 \times 14 / 100 = 2.1 \text{ g of N}$$